

Question	Answer	Marks
6(a)(i)	conversion from a.c. to d.c.	B1
6(a)(ii)	smoothing	B1
6(b)(i)	$A = 12 \text{ V}$	A1
	$B = 2\pi / (20 \times 10^{-3})$ $= 310 \text{ rad s}^{-1}$	A1
6(b)(ii)	full-wave (rectification)	B1
6(b)(iii)	four diodes shown, with correct circuit symbols	B1
	four diodes correctly connected to form a bridge rectifier	B1
6(b)(iv)	$V = V_0 \exp(-t / \tau)$ or $V = V_0 \exp(-t / RC)$ and $\tau = RC$	C1
	$8.0 = 12 \exp(-7.3 \times 10^{-3} / \tau)$	C1
	$\tau = 0.018 \text{ s}$	A1
6(c)	time constant = RC	C1
	$R = (0.018 / 570 \times 10^{-6})$ $= 32 \Omega$	A1

9702/41

Cambridge International AS & A Level – Mark Scheme
PUBLISHED

May/June 2025

Question	Answer	Marks